Laboratories for Universal Negatives

P versus **NP** as a question about the experimental regimes available to a mortal scientist

L. G. Meredith
F1R3FLY.io `lgreg.meredith@f1r3fly.io`

October 2026 — draft 3

Abstract

A mortal scientist is a cost-accounted computation that learns about the computations around it by perceiving them, interacting with them, and reflecting on the code of those it holds. Its evidence is asymmetric: an existential claim about an environment is confirmed by one successful experiment, while a universal claim is only ever refuted. This note asks what that asymmetry costs, and recasts the separation of **P** from **NP**, if there is one, as a statement about it. Fix a bounded specimen and ask the scientist to certify that the specimen cannot reach a designated state. We define an *experimental regime* — a grade of access, an intervention language, a readout and a budget — and show that the regimes split. Regimes restricted to interaction provably cannot certify such universal negatives at polynomial cost: an adversary argument forces 2^d intervened copies on specimens of size $O(d \log d)$. Regimes with reflection can do so if and only if $\mathbf{P} = \mathbf{NP}$, under three stated cost hypotheses. We then take stock of two tools. Fuzzy rho, whose graded communication modulates the code being sent, supplies *knockout* experiments: by choosing the channel on which a specimen is sent, the experimenter chooses a grade that deletes pieces of it, and with stamps, a unit-grade control and pooled readout this gives isolated, attributable interventions. Deleting whole prefixes is a sound intervention and keeps the interaction bound; weakening a synchronisation defeats it. The rho combinators supply a literal combinatorial account of where the complexity lives: we show that the universal negative is **coNP**-complete already for the two atoms $\{\mathsf{m}, \mathsf{sy}\}$, where every run is of linear length, and in **NL** for the join-free atoms $\{\mathsf{fw}, \mathsf{kk}, \mathsf{dd}, \mathsf{q}\}$, so that duplication alone adds no hardness. Hardness lives in synchronisation; the cost of an experimenter lives in contraction. We combine the tools — regimes as points of a product lattice, knockouts on the combinator image, the cost of sharing after Asperti, and a light namespace logic in which the cost hypothesis becomes a type — and close by describing what it would mean for a mortal scientist to live in a world where $\mathbf{P} = \mathbf{NP}$, and in one where it does not.

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1 Aim

The aim of this note is to frame the **P** versus **NP** question as a question about experimental regimes, in the sense of the mortal scientist [17], and to state the problem in that frame in a way a reader can follow without first reading the companion notes.

The mortal scientist is a model of learning in which a learner is a cost-accounted computation in the reflective higher-order (rho) calculus, its environment is other computations, its hypotheses are formulae of a namespace logic, and its experiments are assays whose results are paid for out of a finite token stack. Two features of that model make it a natural setting for the question.

The scientist's evidence is already asymmetric. In the scientist's logic, an existential claim about an environment — some interaction leads somewhere — is confirmed by a finite witness, while a universal claim — no interaction leads there — can be refuted by a single position but never finitely confirmed by running [17, Prop. Asymmetry]. This is the **NP**/**coNP** asymmetry seen from inside a learner. The scientist is never asked to decide a language. It is asked to pay for evidence, and the question that matters to it is what evidence for a universal negative costs.

The scientist's access is already graded. A scientist has exactly three grades of access to its environment: *perception* of the surface, *interaction* by probing, and *reflection* on the code of names it holds [17, §2]. These grades are priced differently, and they are ordered by invasiveness. The separation of **P** from **NP** turns out to be a statement about exactly one of them.

The reframing does three things the standard formulation does not.

(1) It makes the *apparatus* explicit. What an experimenter can do with a specimen — run it once, run it repeatedly, intervene on it, read it — becomes a choice of regime, and the known barriers to proving $\mathbf{P} \neq \mathbf{NP}$ appear as features of the apparatus rather than as facts about proofs.

(2) It makes the *cost model* explicit. The note's central equivalence holds only because the cost model charges for unshared work, and Asperti's observation that $\mathbf{P} = \mathbf{NP}$ "up to sharing" [1] identifies exactly the clause that must not be relaxed.

(3) It gives two existing tools traction. Fuzzy rho [7] supplies controlled experiments in which pieces of the code being sent are knocked out, the experimenter choosing what to delete by choosing where to send. The rho combinators [18] supply a combinatorial account in which one can include or exclude individual atoms and watch the complexity of the scientist's problem change.

Plan. Section 2 assembles the apparatus: the rho calculus, context-labelled bisimilarity, the mortal scientist, and the cost model. Section 3 states the problem in the frame and proves the two results that anchor it. Sections 4 and 5 take stock of fuzzy rho and the rho combinators. Section 6 develops ways of combining them. Section 7 concludes with what each answer to the question would mean for a mortal scientist. Appendix A records the status of each result. Nothing here claims progress on **P** versus **NP** itself; the claim is that this is the right laboratory in which to look at it.

2 The apparatus

2.1 The rho calculus

Definition 2.1 (Syntax)**.** Processes P and names x are given by mutual recursion:

$$P \;::=\; \mathbf{0} \;\mid\; P_1 \mid P_2 \;\mid\; x!(P) \;\mid\; \mathsf{for}(y_1 \leftarrow x_1 \,\&\, \cdots \,\&\, y_n \leftarrow x_n)P \;\mid\; *x \qquad\qquad x \;::=\; @P$$

$x!(P)$ is the *lift*: it quotes P and outputs the name $@P$ on x. The input binds $y_1, \ldots, y_n$ in P; for $n = 1$ we write $\mathsf{for}(\, y \leftarrow x \,)P$, and for $n > 1$ it is a *join*. $*x$ is the *drop*. We write _ for a bound name that does not occur. A process is *closed* if no variable bound by an input occurs free in it.

Definition 2.2 (Name equivalence, structural congruence, reduction)**.** As in [15]: $\equiv_{\mathcal{N}}$ is the least equivalence on names with $@P_1 \equiv_{\mathcal{N}} @P_2$ whenever $P_1 \equiv P_2$, and $@(*x_1) \equiv_{\mathcal{N}} x_2$ whenever $x_1 \equiv_{\mathcal{N}} x_2$; $\equiv$ is the least congruence containing α-equivalence and the abelian monoid laws for $|$ with unit $\mathbf{0}$, names being compared up to $\equiv_{\mathcal{N}}$. Reduction is

$$\frac{x_i \equiv_{\mathcal{N}} x'_i \quad (1 \le i \le n)}{\mathsf{for}(y_1 \leftarrow x_1 \,\&\, \cdots \,\&\, y_n \leftarrow x_n)P \;\mid\; x'_1!(Q_1) \mid \cdots \mid x'_n!(Q_n) \;\rightarrow\; P\{@Q_1/y_1, \ldots, @Q_n/y_n\}}$$

closed under $|$ and $\equiv$. Substitution replaces all names $\equiv_{\mathcal{N}}$-equal to the variable, reaches the object of a lift, does not reach beneath a quote, and performs the drop: $(*x)\{@P/y\} = P$ when $x \equiv_{\mathcal{N}} y$.

This is the calculus of [19] with joins, in the notation of [15]. A name is in *normal form* if no quote in it immediately encloses a drop; rewriting $@(*x)$ to x throughout gives a normal form $\equiv_{\mathcal{N}}$-equal to any name. Name equivalence is decidable in polynomial time, being equality of unordered trees after normalisation.

Three facts about names are used throughout.

Proposition 2.3 (Well-foundedness)**.** *No process P has $@P$ among its names, up to the identity $@(*x) \equiv_{\mathcal{N}} x$.*

Proof sketch. A name occurring in P quotes a process strictly smaller than P, so its quote depth is at most that of P, while $@P$ has quote depth one greater. The only exception is the degenerate $P = *x$, where $@P \equiv_{\mathcal{N}} x$; quote depth in the sense of [14] assigns $@(*x)$ the depth of x and defines it away. □

Consequently a name fresh for P can be manufactured by quoting any term that properly contains P, without a name server [17, 15], and `new` is syntactic sugar. Self-reference is never a matter of a term containing its own name; it can only arise at run time, by communication.

Proposition 2.4 (One quote level per rendezvous)**.** *A reduction adds at most one level of quote depth. Without joins it adds at most one name [15]; an n-ary join adds at most n.*

The *duplicator* $\mathsf{D}(x) \triangleq \mathsf{for}(\, y \leftarrow x\,)(*y \mid x!(*y))$ satisfies $x!(Q) \mid \mathsf{D}(x) \to Q \mid x!(Q)$. Since $@(*y) \equiv_{\mathcal{N}} y$, the duplicator re-sends what it receives *without* raising its quote depth. It is the source of contraction on names, and of replication: $x!(\mathsf{D}(x)) \mid \mathsf{D}(x)$ runs forever.

2.2 Context-labelled transitions

We do not use barbs. Following [15], observation is by contexts.

Definition 2.5 (Context-labelled transitions and bisimilarity)**.** Write $P \xrightarrow{C} P'$ when $C[P] \to P'$, and $P \xRightarrow{C} P'$ when $C[P] \to^* P'$. For an *observer class* $\mathcal{D} \subseteq \mathcal{C}$, a symmetric relation $\mathcal{R}$ is a $\mathcal{D}$-bisimulation if $P \mathrel{\mathcal{R}} Q$ and $P \xrightarrow{C} P'$ with $C \in \mathcal{D}$ imply $Q \xRightarrow{C} Q'$ with $P' \mathrel{\mathcal{R}} Q'$; $\approx_{\mathcal{D}}$ is the largest such. An $\mathcal{D}$-trace of P is a sequence $C_1 \cdots C_k$ of contexts in $\mathcal{D}$ with $P \xRightarrow{C_1} P_1 \cdots \xRightarrow{C_k} P_k$; $P \equiv^{\mathcal{D}}_{\mathrm{tr}} Q$ when they have the same $\mathcal{D}$-traces. Plainly $\approx_{\mathcal{D}} \subseteq \equiv^{\mathcal{D}}_{\mathrm{tr}}$.

Definition 2.6 (Success)**.** Fix a name ok and let $C_{\mathsf{ok}} = [\,\cdot\,] \mid \mathsf{for}(_ \leftarrow \mathsf{ok})\mathbf{0}$. P *can succeed* if $P \xRightarrow{[\,\cdot\,]} P' \xrightarrow{C_{\mathsf{ok}}} P''$ for some P', P'', the displayed step consuming an output on ok.

Success is a context-labelled transition, not a barb.

Remark 2.7 (Holes under a lift). Context-labelled bisimilarity in rho is a congruence for the operators when observations are suitably restricted, and fails to be one for contexts whose hole lies under a lift or a quote, because communication depends on name equality rather than on bisimilarity [28]. The contexts that matter in §3.3 are exactly of that kind.

2.3 The mortal scientist

We recall from [17] only what is used. The premise is that computations are ontologically isolated, so the environment of a computation is other computations, and a world factors across an interaction cut as $W \equiv S \mid E$, a scientist S and its environment E.

Three grades of access. S does not hold E's term. It has exactly three kinds of access, and the calculus offers no fourth [17, §2.2].

grade	instrument	applies to	price
perception	structural predicates	the surface of E	monotone in formula depth
interaction	$\langle K_j \rangle \varphi$	the behaviour of E	one rendezvous per step, and time
reflection	name predicates $@\varphi$	names S holds	a computation on a name; no communication

The modality $\langle K_j \rangle$ is generated by the rewrite rule and is labelled by the redex context: for $W \equiv K_j[\mathsf{for}(\, x \leftarrow n \,)P \mid n!(Q)]$ the label K_j is everything else, so $\exists j.\, \langle K_j \rangle \varphi$ and $\forall j.\, [K_j]\varphi$ quantify over redex positions. The name predicate satisfies $n \models @\varphi$ iff $n = @P$ with $P \models \varphi$; it is namespace logic [20]. Reflection is cheap for the reason predation is not: opening a name one holds is not a passage through a surface.

Assays. A hypothesis installed as the guard of an input is an experiment. The rho variant whose for-comprehension carries a `where` clause [23] realises an *assay* on probe channel p as a complementary pair of guarded receipts,

$$K[\cdots] \mid \mathsf{for}(\, x \leftarrow p \ \texttt{where}\ \varphi(x) \,)\mathit{confirm}!(*x) \mid \mathsf{for}(\, x \leftarrow p \ \texttt{where}\ \neg\varphi(x) \,)\mathit{refute}!(*x),$$

and, when φ is decidable on the received term within the budget, yields one of three outcomes: CONFIRM, REFUTE, or $\perp$ when nothing arrives within the budget [17, Prop. Trichotomy]. A guard may test a name predicate on a received quoted process. So in the `where`-clause variant of rho, reflection is available operationally: a scientist that receives $@E$ can inspect E's code without running it.

The asymmetry. For $\varphi = \exists j.\, \langle K_j \rangle \psi$, CONFIRM is a finite witness, while refuting it requires ruling out every position; for $\forall j.\, [K_j]\psi$ the polarity reverses [17, Prop. Asymmetry]. A scientist with a finite budget can therefore test only a fragment bounded in modal depth and fixed-point unfolding, and *the budget is the inductive bias.*

The game. Tokens are conserved and never minted. A mortal computation exposes its stack as a message on a *metabolic channel*; a harvest is an ordinary receipt on that channel, and it ends the prey's computational life [17, §7]. Under *capability gating* a predator must derive a metabolic name from structure it can perceive, and derivation is real work. Two consequences matter here.

- *One specimen, one measurement.* After a destructive assay no further assay may be run against the specimen [17, Prop. One specimen]. This is the linear tier of §3.4.
- *Concealment is a price, not a wall.* Because names are quoted processes and `new` is sugar, nothing is secret; it is only expensive to find [17, Rem. Restriction is sugar].

2.4 Cost

Costs are read in cost-accounted rho, whose phlogiston charges communication [6]. We need three properties of the charge, stated as hypotheses because they are properties of a cost model, not of the calculus.

Hypothesis 2.8 (Simulation)**.** There is a polynomial q such that, for every deterministic Turing machine M, there is a rho process $R_M(x)$ which, given an input word w as data on x, runs determinately to a representation of $M(w)$ at phlogiston at most $q(t_M(w) + |w|)$, where $t_M(w)$ is the running time.

Hypothesis 2.9 (Accounting)**.** Each communication — including the graded communications of §4 — is charged at least 1 plus the total size, *unshared*, of the transmitted terms; and every run of phlogiston c from a term of size n, including its guard evaluations and the copying performed by substitution into drops, can be simulated by a deterministic Turing machine in time polynomial in $c + n$.

Hypothesis 2.10 (Reflection)**.** In the reflection tier, guards are *structural*: boolean combinations of term-former connectives and name predicates, without behavioural modalities or fixed points. Evaluating such a guard of size s on a held name quoting a term of size n costs phlogiston at least 1 and at most polynomial in $s + n$.

The second clause of Hypothesis 2.9 is stated as an assumption, not derived, because it must cover steps that communication charges alone do not meter. Hypothesis 2.10 excludes guards such as $\Diamond\langle C_{\mathsf{ok}}\rangle\top$, which the `where` clause of [23] would otherwise admit and which would decide the problem of §3 in one unpriced step.

The word *unshared* in Hypothesis 2.9 carries the whole weight of §5.4. Hypothesis 2.10 is a constraint on the pricing schedule of reflection; the sensing schedules used in the companion game note [8], exponential in depth, apply to perception across the cut, not to reading a name one holds.

3 The problem in the frame

3.1 Bounded specimens and the universal negative

The scientist's question is about a specimen whose behaviour is bounded, so that every run is short and the only difficulty is in the branching.

Definition 3.1 (Flat specimens)**.** A process is *flat* if it is closed, contains no drop (not even beneath a quote), and no bound name occurs in subject position. In a flat process each communication consumes an input prefix not under a lift and creates none, since no transmitted term is ever run; so every run of a flat P has length at most $|P|$. A *race* in P is a set of input prefixes competing for one output; a *resolution* of P assigns a winner to each race.

Definition 3.2 (The universal negative)**.** ABSENCE is the problem: given a flat P, decide that P *cannot* succeed. In the scientist's logic this is the universal formula $\neg\,\exists\,\mathrm{run}.\,\Diamond\langle C_{\mathsf{ok}}\rangle\top$ — a claim that no position, at any point of any run, offers ok.

Proposition 3.3. ABSENCE *is* **coNP***-complete.*

Proof. It is in **coNP**: a successful run has length at most $|P|$ and is a certificate of success, checkable in polynomial time because name equivalence is. For hardness, reduce from CNF unsatisfiability. Given variables $v_1,\dots,v_d$ and clauses $c_1,\dots,c_m$, fix pairwise distinct closed names $b_i, t_{ij}, f_{ij}, e_j, \mathsf{ok}$ and let

$$\mathit{Race}_i \;=\; b_i!(\mathbf{0}) \;\mid\; \mathsf{for}(_ \leftarrow b_i)\textstyle\prod_{j:v_i\in c_j} t_{ij}!(\mathbf{0}) \;\mid\; \mathsf{for}(_ \leftarrow b_i)\prod_{j:\neg v_i\in c_j} f_{ij}!(\mathbf{0}),$$

let Clause_j compose $\mathsf{for}(_ \leftarrow t_{ij})e_j!(\mathbf{0})$ for each positive and $\mathsf{for}(_ \leftarrow f_{ij})e_j!(\mathbf{0})$ for each negative occurrence in c_j, and put $P_\varphi = \prod_i \mathit{Race}_i \mid \prod_j \mathit{Clause}_j \mid \mathsf{for}(_ \leftarrow e_1 \,\&\, \cdots \,\&\, _ \leftarrow e_m)\,\mathsf{ok}!(\mathbf{0})$. Each race fixes one truth value; a clause signals only if one of its literals was made true; the join fires only if every clause signalled. P_φ can succeed iff φ is satisfiable. □

So the specimen's universal negative is as hard as any **coNP** problem, and the hardness is entirely in the races: no run is long. The scientist who could choose the resolution would need one experiment to confirm success. Confirming its absence is the problem.

3.2 The same problem, stated by bisimulation

The asymmetry has a bisimulation-native form, which is how it was first reached and which shows that the question is native to the calculus rather than imported.

Definition 3.4 (Explicit bounded processes)**.** An *explicit bounded process* is a finite acyclic labelled transition system $G = (S, \Sigma, \to, s_0)$, given as a list of transitions. G is *determinate* if every state has at most one a-successor for each a. $\sim$ is strong bisimilarity and $\equiv_{\mathrm{tr}}$ trace equivalence.

Construction 3.5 (Encoding)**.** With pairwise distinct closed names $\mathsf{go}_s, \mathsf{l}_s$ for states and a for labels,

$$\mathcal{E}(s) \;=\; \mathsf{l}_s!(\mathbf{0}) \;\mid\; \prod_{(s,a,t)\in\to} \mathsf{for}(_ \leftarrow a \,\&\, _ \leftarrow \mathsf{l}_s)\,\mathsf{go}_t!(\mathbf{0}),$$

$$\mathcal{E}(G) \;=\; \mathsf{go}_{s_0}!(\mathbf{0}) \;\mid\; \prod_{s\in S} \mathsf{for}(_ \leftarrow \mathsf{go}_s)\mathcal{E}(s),$$

observed by $\mathcal{D}_\Sigma = \{[\,\cdot\,] \mid a!(\mathbf{0}) : a \in \Sigma\}$. The lock l_s lets one outgoing transition of s fire; a shared state is encoded once. Communications on go-names are administrative and deterministic; taking transitions to be principal steps that consume the offered $a!(\mathbf{0})$, followed by administrative normalisation, $s \sim t$ iff $\mathcal{E}(s) \approx_{\mathcal{D}_\Sigma} \mathcal{E}(t)$, and likewise for traces.

On explicit bounded processes, $\sim$ is decidable in polynomial time [24] (**P**-complete on finite systems in general [5]), while $\equiv_{\mathrm{tr}}$ is **coNP**-complete, by way of the inequivalence of regular expressions built from union and concatenation only [27], with an end marker to handle prefix-closed traces. Branching time is cheap; linear time is not.

Definition 3.6 (Normaliser)**.** A *normaliser* is a polynomial-time map Φ on explicit bounded processes with $G \equiv_{\mathrm{tr}} H \iff \Phi(G) \sim \Phi(H)$. It is *faithful* if $\Phi(G) \equiv_{\mathrm{tr}} G$.

Theorem 3.7. P = NP *if and only if a normaliser exists, and then a faithful one exists. Equivalently: the projection $\pi : B/\!\sim \to B/\!\equiv_{\mathrm{tr}}$ has a polynomial-time section.*

Proof. ($\Leftarrow$) Decide $G \equiv_{\mathrm{tr}} H$ by testing $\Phi(G) \sim \Phi(H)$; this puts a **coNP**-complete problem in **P**. ($\Rightarrow$) If **P** = **NP** the polynomial hierarchy collapses to **P**. Let $m(G)$ be the least size of a process trace-equivalent to G, and let $\Phi(G)$ be the lexicographically least such process of size $m(G)$, over states numbered $0, \ldots, m$ and the labels occurring in traces of G. Both depend only on the trace class. Its bits are found by Σ_2^p questions (there is a G' extending a given prefix such that, for all traces up to the depth bound, membership agrees), hence in polynomial time. Then Φ is faithful, equal trace classes give equal normal forms, and $\Phi(G) \sim \Phi(H)$ gives $\Phi(G) \equiv_{\mathrm{tr}} \Phi(H)$ and so $G \equiv_{\mathrm{tr}} H$. □

Proposition 3.8 (Faithful determinate normalisers do not exist)**.** *There are G_n with $|G_n| = O(n^2)$ such that every determinate $H \equiv_{\mathrm{tr}} G_n$ has at least 2^n states.*

Proof. Let G_n have the traces of the words of $L_n = \{uv \in \{0,1\}^{2n} : u \neq v,\ |u| = |v| = n\}$, each followed by ok: it guesses a position, records a bit, counts n further and checks the bit differs, with states indexed by position, guessed index and bit. For determinate $H \equiv_{\mathrm{tr}} G_n$ each $u \in \{0,1\}^n$ leads to a unique state h_u, and for $u \neq u'$ the trace $u'\mathsf{ok}$ is available from h_u but not from $h_{u'}$. This is the blow-up of determinisation [21, 22]. □

Without faithfulness a determinate normaliser exists iff **P** = **NP**: spell out the bits of the canonical form of Theorem 3.7 as a straight line. So the type of a normaliser is a *section of the coarsening projection*, choosing one bisimilarity class inside each trace class, and the question is whether such a choice can be made cheaply. A choice that lands in determinate processes is provably expensive.

3.3 Internalising the experimenter

Inside the calculus the experimenter is a process. We write it $[\![\Phi]\!](x)$, a fixed term with a free name x on which it expects its specimen, and the configuration is

$$[\![\Phi]\!](x) \ \mid\ x!(E) \ \ \to^* \ \ \text{verdict}.$$

In context-labelled terms the experimenter *is* the context $C_\Phi = [\![\Phi]\!](x) \mid x!([\,\cdot\,])$, whose hole lies under a lift. By Remark 2.7 this is where bisimilarity fails to be a congruence: invariance of the verdict under $\equiv_{\mathrm{tr}}$ is a burden the experimenter carries, not a property it inherits.

Lemma 3.9 (What a received name is good for)**.** *In $\mathsf{for}(\,y \leftarrow x\,)R$ every free occurrence of y in R is the subject of an input or output or the argument of a drop. So, in pure rho, once $[\![\Phi]\!](x)$ has received $@E$ it can (i) run E, by a drop; (ii) use $@E$ as a channel, which tests it for name equality against names it or its environment constructs; or (iii) forward it, deferring (i) or (ii).*

Proof. Names occur only as subjects, binders, under drop, and beneath quotes, where they are not free. □

So in pure rho every test on a specimen's code is an interaction. What separates a held specimen from a live one is *resource*: interaction with a live E can use it up, while a name can be dropped as often as the experimenter pays for. The `where`-clause variant adds the scientist's third grade: a guard may evaluate a name predicate on $@E$ without running it.

Example 3.10 (Why one experiment is not enough). Let $G = a.(b + c)$ and $H = a.b + a.c$, so $G \equiv_{\mathrm{tr}} H$ and $G \not\sim H$. An experimenter interacting with a live $\mathcal{E}(H)$ learns nothing until it offers a, and the join that consumes the offer also chooses the branch. The losing branch survives only as a dead join on a lock no observer in $\mathcal{D}_\Sigma$ can name. A faithful normal form must have both ab and ac, and a single experiment cannot recover the trace it did not see.

3.4 Access tiers are structural rules

Quotation gives the specimen the exponential of linear logic: $@E$ behaves as $!E$, the drop as dereliction, the duplicator as contraction. The scientist's grades of access become tiers distinguished by which of these rules the experimenter may use, and whether it may read.

Definition 3.11 (Tiers)**.** (a) *Linear* (destructive assay): the received name is dropped at most once, as in the exactly-once input $\mathsf{for}(y \multimap x)P$. No contraction. Type $B \multimap B$.

(b) *Interaction* (black box): the received name is used only under drop, and forwarded only to receivers that do likewise; the experimenter's contexts mention, of the specimen's names, only a declared public interface. Contraction is allowed; reading and name guessing are not. Type $!B \multimap B$ with ! extensional.

(c) *Reflection* (white box): as (b), and in addition the experimenter may evaluate name predicates on the received name — on the quoted code, which does not run while it is read. Type $@B \to B$, with @ an intensional box whose eliminators may inspect code.

A grading (§4) applied to a specimen in transit belongs to the interaction tier only if it is injective and blind to the structure of the specimen's names; a grading that computes on that structure is a reflection-tier act, whoever chooses it (§6.3).

Conjecture 3.12 (Drop-only usage is extensional)**.** *If the hole of C lies under a lift and the corresponding name is used drop-only, then $P \approx_{\mathcal{D}} Q$ implies $C[P] \approx_{\mathcal{D}} C[Q]$ for suitable $\mathcal{D}$: @ restricted to dereliction and contraction is* !.

3.5 Experimental regimes

Definition 3.13 (Regime)**.** An *experimental regime* is a tuple $\mathcal{R} = (A, I, O, c)$: an access tier A; an intervention language I, a set of partial resolutions the experimenter may impose on a copy of the specimen (realised, where they are realised at all, by the knockouts of §4); a readout discipline O (single, pooled, or graded, read through the assay trichotomy); and the cost model c of §2.4. A *copy* is released by one communication that applies its signature and intervention, and is identified with it. An *experiment* is a finite pool of copies run against test contexts allowed by A. $[\![\Phi]\!](x)$ *exploits* $\mathcal{R}$ for Absence if, for every flat E, the configuration $[\![\Phi]\!](x) \mid x!(E)$ performs only experiments of $\mathcal{R}$, runs at phlogiston polynomial in $|E|$, and reports, independently of the scheduler, whether E can succeed.

Lemma 3.14 (Copies cost)**.** *Under Hypothesis 2.9, a run of phlogiston c releases at most c copies.*

Proof. Each copy is its own communication, charged at least 1. □

Unstamped duplicates are cheaper — one communication can deliver $@E$ to a body with many drops — but they share channels, are neither isolated nor attributable, and carry no intervention; under the adversary below they contribute nothing.

Definition 3.15 (Interaction regime)**.** $\mathcal{R}$ is an *interaction regime* if its tier is (a) or (b) and the outcome of each experiment on E is determined by the experiment and by whether each copy, under its intervention, succeeds at its stamped interface. Interventions are granted as a primitive over declared race ports. This is more than such an experimenter could otherwise do: the listeners of a race share a subject, so separating them needs a grade that discriminates on the structure of their continuations, which belongs on the reflection side (§6.3). Likewise success or failure of a copy is granted as an observation, although by [17, Prop. Asymmetry] a failure is in general only $\bot$; flat copies are bounded, and we grant quiescence detection. Both grants only strengthen what follows. Races an intervention leaves unresolved are resolved by a scheduler outside the experimenter's control, and correctness is required under every scheduler.

Theorem 3.16 (Interaction cannot certify universal negatives)**.** *For each d there are flat $E_\bot$ and E_σ, $\sigma \in \{0,1\}^d$, of size $O(d \log d)$, such that $E_\bot$ cannot succeed, each E_σ can, and every $[\![\Phi]\!](x)$ that, using an interaction regime, reports correctly and scheduler-independently on all of them releases at least 2^d intervened copies on $E_\bot$; under Hypothesis 2.9 its phlogiston there is at least 2^d.*

Proof. Let $Race_i = b_i!(\mathbf{0}) \mid \mathsf{for}(_ \leftarrow b_i)e_i^0!(\mathbf{0}) \mid \mathsf{for}(_ \leftarrow b_i)e_i^1!(\mathbf{0})$, and put

$$E_\sigma = \prod_{i \le d} Race_i \;\mid\; \mathsf{for}(_ \leftarrow e_1^{\sigma_1} \,\&\, \cdots \,\&\, _ \leftarrow e_d^{\sigma_d})\, \mathsf{ok}!(\mathbf{0}),$$

$$E_\bot = \prod_{i \le d} Race_i \;\mid\; \mathsf{for}(_ \leftarrow e_1^0 \,\&\, \cdots \,\&\, _ \leftarrow e_d^0 \,\&\, _ \leftarrow z)\, \mathsf{ok}!(\mathbf{0}),$$

with z a name nothing sends on and all names distinct (quoted binary numerals give the size bound). E_σ succeeds exactly under the resolution σ; $E_\bot$ never does. A copy with intervention τ fails on $E_\bot$, and fails on E_σ unless τ is the full assignment σ: if τ disagrees with σ the join cannot fire, and if τ leaves a race open a scheduler resolving it against σ makes the copy fail. Fix such a scheduler. Every experiment then returns the same outcome on E_σ as on $E_\bot$ unless it contains a copy with full intervention σ. Run $[\![\Phi]\!](x)$ on $E_\bot$; it reports failure. If some σ was never applied, then on E_σ every experiment returns what it returned on $E_\bot$, the run makes the same adaptive choices by induction, and it wrongly reports failure. So all 2^d full interventions were applied, and Lemma 3.14 gives the cost bound. □

Corollary 3.17. *No interaction regime is exploitable for* Absence*; in particular the linear regime — one specimen, one measurement — is not.*

Remark 3.18 (Why this says nothing about **P** versus **NP**). With reflection the family is trivial: $[\![\Phi]\!]$ reads the final join and sees either the conjunct on z or the hidden σ. The bound is purely about access. It is a query lower bound for unstructured search, the in-calculus analog of the

separating half of the Baker–Gill–Solovay theorem [4]: an interaction-only scientist computes with the specimen as an oracle. We give no analog of the collapsing half, so the correspondence with relativisation is an analogy.

Theorem 3.19 (Reflection is exactly as hard as **P** versus **NP**)**.** *Under Hypotheses 2.8–2.10,* $\mathbf{P} = \mathbf{NP}$ *if and only if some regime is exploitable for* Absence*; when one is, a reflection regime is.*

Proof. ($\Rightarrow$) Absence $\in \mathbf{P}$, decided by some M. A reflection regime reconstructs E from $@E$ by adaptive structural guards: at each node of the term it asks which term former is at the root and compares subject names against names it has already reconstructed, a polynomial number of guards each of polynomial size, re-sending $@E$ to itself between queries (sketched). By Hypothesis 2.10 this costs polynomial phlogiston; it then runs R_M on the reconstruction (Hypothesis 2.8). No copy of E runs, so the report is scheduler-independent. ($\Leftarrow$) The report of an exploiting $[\![\Phi]\!]$ is scheduler-independent, so simulating any one schedule computes it; the run costs polynomial phlogiston, so by Hypothesis 2.9 a deterministic machine simulates it in polynomial time, and Absence $\in \mathbf{P}$. □

Corollary 3.20 (The problem, in the frame)**.** *Under the hypotheses of §2.4:*

> $\mathbf{P} \neq \mathbf{NP}$ *if and only if no mortal scientist, in any experimental regime, can certify uniformly, at cost polynomial in the size of the specimen, that a flat specimen cannot reach a designated state.*

Interaction regimes provably cannot (Theorem 3.16). The remaining content of $\mathbf{P} \neq \mathbf{NP}$ *is therefore a statement about reflection: that holding a specimen's code affords no laboratory for its universal negatives. A proof of* $\mathbf{P} = \mathbf{NP}$ *would exhibit one.*

Theorem 3.19 is a restatement, not a technique. What the frame adds is a vocabulary in which the parts of the restatement — the tier, the intervention language, the readout, the cost model — can be varied one at a time. The next two sections take stock of the tools for varying them.

4 Tool I: fuzzy rho, and knocking out pieces of the code

Repetition alone yields only existential evidence. If a copy of the specimen succeeds, the specimen can succeed; if it fails, the scheduler may simply have resolved a race the other way. A biologist facing the same problem knocks genes out: removes one component, holds the rest fixed, and compares with a control. Fuzzy rho supplies the corresponding apparatus, because in it the communication rule itself modulates the code being sent, and the experimenter controls the modulation by choosing where to send.

4.1 Graded communication

Fuzzy rho [7] adds an action of grades on processes and a comparison of interaction surfaces,

$$(- \star -)\colon A \times \mathsf{Proc} \to \mathsf{Proc}, \qquad (- \circ -)\colon \mathsf{N} \times \mathsf{N} \to A,$$

with A a quantale and a monotone measure $\mu : \mathsf{Proc} \to A$, and replaces strict channel equality by a graded rule

$$\mathsf{for}(\, y \leftarrow x \,)R \;\mid\; u!(Q) \;\to\; R\{@((u \circ x) \star Q)/y\}.$$

The *order axiom* requires $a_1 \leq a_2 \Rightarrow \mu(a_1 \star P) \leq \mu(a_2 \star P)$: a larger grade transmits at least as much. The action is a parameter of the calculus, constrained only by this axiom.

The minimal model of [7] takes grades to be names, $u \circ x = @(\overline{u} \mid \overline{x})$ (where $\overline{z}$ is the process quoted by z), and an action that re-channels every free name of the transmitted term,

$$(u \circ x) \star Q = Q\{@(\overline{u} \mid \overline{x} \mid \overline{z})/z : z \in \mathrm{fn}(Q)\}.$$

This *accumulator* never deletes: it stamps every channel of Q with the match signature of the redex, and the order axiom holds in it modulo a congruence side-condition [7]. Its dual, the *attenuator*, replaces $\mid$ by the multiset meet $\wedge$, so that channels can collapse. Neither deletes, but the calculus does not require an action to preserve terms, and other models can.

4.2 Knockouts

In fuzzy rho any input may react with any output, at the grade that compares their channels. That is the experimenter's dial. To knock a piece out of Q before $\mathsf{for}(\, y \leftarrow x \,)R$ receives it, the experimenter sends Q not on x but on a channel u chosen so that $(u \circ x) \star Q$ is Q with that piece removed. The mismatch between u and x is the treatment.

Three conditions on a model make this an apparatus.

(K1) *A deleting action.* Some grades remove parts of the term they act on.

(K2) *A unit control.* $x \circ x = \mathbf{1}$ and $\mathbf{1} \star Q = Q$, so a copy sent on the receiver's own channel is the untreated baseline. The minimal model fails this: there $x \circ x = @(\overline{x} \mid \overline{x})$, a watermark rather than the identity.

(K3) *A bottom.* $u \circ x = \bot$ for most mismatched pairs, and the rule does not fire at $\bot$. Otherwise a copy's own internal communications are graded too, and it ceases to be a faithful copy.

Construction 4.1 (The deletion model)**.** Call a *piece* of Q either a $\mid$-prime component of Q or of the body of one of its prefixes — a prefix together with its continuation — not beneath a lift or a quote; or a single binding $y_i \leftarrow x_i$ of a join. A grade is $\bot$ or a set Φ of structural formulae, and $\Phi \star Q$ deletes every piece of Q satisfying some formula in Φ, the formulae being evaluated on the pieces of Q itself and the outermost deletion taking precedence (a deleted prefix piece becomes $\mathbf{0}$; a deleted binding is dropped from its join). Order grades by reverse inclusion, with $\bot$ adjoined below, and take union as product; since union distributes over intersections this is a quantale with unit $\mathbf{1} = \varnothing$. The comparison is $x \circ x = \mathbf{1}$; $u \circ x = \Phi$ when u is an *experimental channel* for x carrying Φ — a name built from $\overline{x}$, a code for Φ, and an unforgeable component held by the experimenter; and $\bot$ otherwise. Since $\Phi_1 \supseteq \Phi_2$ deletes every position Φ_2 deletes, $\Phi_1 \star Q$ is obtained from $\Phi_2 \star Q$ by deleting further pieces. So the order axiom holds with μ taking a term to its quotation ordered by deletion of pieces, just as the minimal model of [7] takes $\mu(P) = @P$ under a precongruence on names: a smaller grade deletes more.

As stated in [7] the measure lands in the grade quantale; here it lands in names ordered by deletion, as in the minimal model, where grades and names coincide. Whether a single quantale can carry both is part of open problem 7.

The unforgeable component ensures that no two of the specimen's own channels form an experimental pair, so inside a copy only exact matches fire and the copy runs as ordinary rho (K3). A grade made only of formulae picking out prefix pieces is a *prefix-deleting* grade. One that picks out bindings is a *weakening* grade, because it relaxes a synchronisation. The distinction matters.

Lemma 4.2 (Prefix deletion is run-subset)**.** *Let Φ be prefix-deleting. If $\Phi \star Q \to^* R$, then $Q \to^* R \mid D$ for some D; in particular if $\Phi \star Q$ can succeed, so can Q, by a run that resolves every race the same way.*

Proof. A deleted prefix piece is a parallel component of Q or of a body in Q, so Q differs from $\Phi \star Q$ by components that are released as their enclosing prefixes fire. By induction on the run, each term reached from $\Phi \star Q$ is matched by a term reached from Q that adds the released pieces in parallel, with the same substitutions applied. Reduction is closed under parallel composition, so extra components never disable a step. Pieces beneath a lift are excluded because a transmitted term may later be run or used as a name, and deleting inside it would change what the receiver does. □

So a prefix-deleting knockout cannot manufacture success: positive evidence from a knocked-out copy is evidence about the specimen. A weakening grade has no such guarantee. Dropping a binding from a join lets it fire with less than the specimen requires, which creates runs the specimen does not have; §6.3 shows what that buys.

4.3 Isolation and attribution: the stamp

Copies run side by side must not interfere, and each observation must be traceable to its copy. The accumulator's stamp does this, and can be composed with a deletion grade: the experimenter's grade for a copy is a deletion set together with a signature.

Definition 4.3 (Signature, stamp)**.** A *signature* is a closed process S having, as a $|$-prime component, an inert input $K_S = \mathsf{for}(_ \leftarrow k_S)\mathbf{0}$ on an unforgeable name k_S occurring nowhere else. The *stamp* θ_S sends each name z in normal form to $@(S \mid \overline{z})$, and $S \star P$ applies θ_S to the names of P.

Lemma 4.4 (Stamps are injective)**.** $\theta_S(z_1) \equiv_{\mathcal{N}} \theta_S(z_2)$ *implies* $z_1 \equiv_{\mathcal{N}} z_2$.

Proof. $S \mid \overline{z}$ has K_S as a component, so it is never $\equiv$ to a lone drop and only the first clause of $\equiv_{\mathcal{N}}$ applies. Up to $\equiv$ a process is a finite multiset of $|$-prime components, and the free commutative monoid is cancellative. □

Lemma 4.5 (Distinct signatures are disjoint)**.** *If the names k_S are globally unique, $k_{S_1} \not\equiv_{\mathcal{N}} k_{S_2}$, and z_1, z_2 are names of a specimen that mentions neither, then $\theta_{S_1}(z_1) \not\equiv_{\mathcal{N}} \theta_{S_2}(z_2)$.*

Proof. If $S_1 \mid \overline{z_1} \equiv S_2 \mid \overline{z_2}$, the prime component K_{S_1} occurs on the right, in S_2 or in $\overline{z_2}$, neither of which mentions k_{S_1}. □

Copies with distinct signatures therefore share no channel and cannot interfere, and every observation carries the signature of the copy that made it. The experimenter knows each signature, so it can compute the stamped interface on which to listen.

4.4 Fidelity: inertness and equivariance

A stamped copy is useful only if it is a faithful copy. If E contains $*z$, the copy contains $*\theta_S(z)$, which runs $S \mid \overline{z}$; signatures must be inert. And flat substitution stamps the names a term has when it is sent, not the names it computes while it runs.

Example 4.6 (Flat stamping is not equivariant). Let $P = a!(\mathbf{0}) \mid \mathsf{for}(\, y \leftarrow a\,)y!(\mathbf{0}) \mid \mathsf{for}(\, _ \leftarrow @\mathbf{0}\,)\mathsf{ok}!(\mathbf{0})$. Then P receives $y := @\mathbf{0}$, outputs on $@\mathbf{0}$, and succeeds. In $S \star P$ the names a, ok and $@\mathbf{0}$ are stamped, but the transmitted $@\mathbf{0}$ is synthesised from the object of a lift that has no names, and is not. The copy outputs on $@\mathbf{0}$ while its listener waits on $\theta_S(@\mathbf{0}) \equiv_{\mathcal{N}} @S$, and cannot succeed.

Definition 4.7 (Experimental grading)**.** A grading is *experimental* if its signatures are signatures in the sense above, pairwise distinct across copies (E1); inert (E2); and the stamp is *equivariant*: $P \to P'$ implies $S \star P \to S \star P'$, and every reduction of $S \star P$ is of this form (E3).

We read (E3) as a run-time analog of the binder transport law $a \star [y]P = [y](a' \star P)$ of [7]: the copy runs inside the stamp. It fails in general, by Example 4.6. Under (E1)–(E3) the stamp is a bisimulation between P and $S \star P$ with observations on z matched to observations on $\theta_S(z)$. Flat specimens synthesise no channel names and contain no drops, so flat stamping is experimental on them.

4.5 Intervention, control, dose and readout

Intervention. Deleting every losing listener of a race resolves it: this is Pearl's do$(\cdot)$ [25], performed by choosing the channel on which the specimen is sent. A *full intervention* resolves every race, and under it a flat copy is confluent. By Lemma 4.2, if the copy succeeds, the specimen succeeds under that resolution. The listeners of a race share a subject, so a grade that deletes one and spares the other must discriminate on the structure of their continuations. The experimenter never inspects the code; the comparison does it on the experimenter's behalf. That is legitimate, but it places a structurally selective grade on the reflection side of the ledger (§6.3).

Control. A copy sent on the receiver's own channel is received at the unit grade, which acts as the identity (K2): the untreated baseline.

Dose. The order axiom says a larger grade transmits at least as much, as measured by μ; in the deletion model, a larger grade deletes less. Read as a dose–response law it licenses inference from graded treatments, but only for outcomes monotone in that measure, which is a further assumption.

Readout. A collapsing grade applied to the experimenter's own channels merges the success channels of a pool onto one name, so that one readout reports whether any member succeeded. Given quiescence detection, a negative pooled readout certifies failure for every member; without it, a negative readout is $\perp$. Pooling saves readouts, not runs.

What knockouts buy. Under a full intervention, and given quiescence detection, a failure is conclusive for that intervention. Knockouts convert demonic scheduling into the experimenter's choice: success of a specimen is the existence of an intervention under which a copy succeeds — an **NP** witness, chosen rather than granted — and failure is failure under every intervention. The universal negative becomes a problem of *experimental design*: can polynomially many knockout experiments cover the space of resolutions?

5 Tool II: the rho combinators, and where the complexity lives

The rho combinators [18] translate rho into a name-free calculus of atoms, each doing one job. Including or excluding atoms gives a lattice of calculi [18, §9], following Yoshida [29]. Because each atom isolates one structural operation, the lattice is a literal map of where the complexity of the scientist's problem lives.

5.1 Atoms and rules

Atoms take names; payloads are written as names in presentation A of [18]. The rules are:

$\mathsf{dd}(a,b,c) \mid \mathsf{m}(a,v) \to \mathsf{m}(b,v) \mid \mathsf{m}(c,v)$	distribute
$\mathsf{kk}(a) \mid \mathsf{m}(a,v) \to \mathbf{0}$	kill
$\mathsf{fw}(a,b) \mid \mathsf{m}(a,v) \to \mathsf{m}(b,v)$	forward
$\mathsf{bl}(a,b) \mid \mathsf{m}(a,v) \to \mathsf{fw}(v,b)$	bind left
$\mathsf{br}(a,b) \mid \mathsf{m}(a,v) \to \mathsf{fw}(b,v)$	bind right
$\mathsf{sy}(a,b,c) \mid \mathsf{m}(a,v) \to \mathsf{fw}(b,c)$	synchronise
$\mathsf{fw}(a,b) \mid \mathsf{q}(a,p) \to \mathsf{m}(b,@p)$	quote
$\mathsf{ev}(a) \mid \mathsf{m}(a,@p) \to p$	evaluate
$\mathsf{cons}_A(a_1,\ldots,a_n,f) \mid \mathsf{m}(a_1,v_1) \mid \cdots \mid \mathsf{m}(a_n,v_n) \to \mathsf{m}(f,@A(v_1,\ldots,v_n))$	construct

with m the message, q a store that only a forwarder can read, and one constructor per atom shape. The translation from rho is linear in size up to a logarithmic factor from its second phase [18, Prop. linear]. We assume its run-time overhead is polynomial, which the source states for the direction from rho to the combinators; the results of §5.3 are proved directly on combinator terms and do not depend on transfer.

Three invariants of [18] carry the lattice.

- *Name closure.* Every name in a reduct lies in the closure of the term's names under the shapes of the constructors present.
- *Weighted atom count.* Weighting $\mathsf{bl}, \mathsf{br}, \mathsf{sy}$ at 2 and other non-message atoms at 1, every rule but ev strictly decreases weight. So without ev every run has length at most twice the number of non-message atoms.
- *Message count.* dd is the only rule that increases the number of messages without opening a payload.

The constructor axis is a Boolean lattice; ev is a strict edge; and the conjecture that the calculus without dd is strongly normalising even with ev and all constructors is open [18, Conj. dup].

5.2 The atoms are structural rules

atom	linear-logic reading	role
dd	contraction	copies a message
kk	weakening	discards a message
ev	dereliction (counit)	opens a name by running it
cons_A	promotion of computed names	builds a name upward
$\mathsf{fw}, \mathsf{bl}, \mathsf{br}, \mathsf{sy}$	axiom links and cut wiring	moves and joins

One operation is missing: a destructor of names that descends without running. We add it as a family

$$\mathsf{m}(a, @A(v_1, \ldots, v_n)) \mid \mathsf{uncons}_A(a, b_1, \ldots, b_n) \;\to\; \mathsf{m}(b_1, v_1) \mid \cdots \mid \mathsf{m}(b_n, v_n),$$

presented as a case split over shapes so that a mismatch does not create a race. It is the combinator form of the observer extension's probes [16] and of the scientist's reflection. By Proposition 2.3, a specimen's internal names lie below $@E$ in the quote-depth order: constructors climb, ev descends by running, and uncons descends without running. Reading is exactly downward motion that does not execute.

5.3 Where the hardness of universal negatives lives

Run Absence on combinator terms: decide that no run delivers a message at ok.

Proposition 5.1 (Join-free atoms are easy)**.** *For terms over* m *and atoms from* $\{\mathsf{fw}, \mathsf{kk}, \mathsf{dd}, \mathsf{q}\}$*,* Absence *is in* **NL**.

Proof. Each rule consumes one message and one atom and produces only messages; no atom is created. So the ancestry of a message at ok is a chain of single-input steps, which is a path in the static graph with an edge $a \to b$ for each $\mathsf{fw}(a, b)$ and edges $a \to b$, $a \to c$ for each $\mathsf{dd}(a, b, c)$ (a q is a message that only fw may move). A simple path uses distinct atoms, and no other message can help or hinder. So reachability of ok is graph reachability, and its complement is in $\mathbf{coNL} = \mathbf{NL}$. □

Theorem 5.2 (Synchronisation is enough)**.** Absence *is* **coNP***-complete for terms over* $\{\mathsf{m}, \mathsf{sy}\}$*, and hence for* $\{\mathsf{m}, \mathsf{fw}, \mathsf{sy}\}$*, although every run has length at most twice the number of atoms.*

Proof. Membership: runs are linear by the weighted atom count. Hardness: read $\mathsf{fw}(a, b)$ as the one-shot rule $a \multimap b$ and $\mathsf{sy}(a, b, c)$ as the one-shot rule $a \otimes b \multimap c$ — a message at a creates $\mathsf{fw}(b, c)$, which then needs a message at b. Given a CNF over $v_1, \ldots, v_n$:

- for each occurrence of a literal in clause j, a resource message $\mathsf{m}(R)$ and the rule $\mathsf{fw}(R, L_j)$;
- for each variable x, one message $\mathsf{m}(X_x)$ racing between $\mathsf{fw}(X_x, T_0)$ and $\mathsf{fw}(X_x, F_0)$;
- a T-chain $\mathsf{sy}(T_{i-1}, R_i, T_i)$ over the resources of the *negative* occurrences of x, ending in $\mathsf{fw}(T_k, G_x)$, and an F-chain over the positive ones;
- a chain of sy joining all G_x and all L_j and ending at ok.

Only one chain per variable completes, since there is one X_x, and completing it consumes every resource of the opposite literal; a clause can then be signalled only by a literal consistent with the chosen assignment. So ok is reachable iff the CNF is satisfiable, and the term is linear in its size. A static $\mathsf{fw}(a, b)$ is $\mathsf{sy}(z, a, b) \mid \mathsf{m}(z)$ with z fresh. □

The construction over $\{\mathsf{m}, \mathsf{fw}, \mathsf{sy}\}$ was checked by exhaustive state-space search against a truth-table SAT check on 920 random CNFs, with complete agreement: 800 instances with up to three variables and four clauses (seeds 1 and 7), and 120 with two variables and three to five clauses (seed 5); 764 satisfiable and 156 unsatisfiable in all. The reduction to $\{\mathsf{m}, \mathsf{sy}\}$ rests on the confluence of $\mathsf{sy}(z, a, b) \mid \mathsf{m}(z)$, which has a single consumer and adds one step per forwarder.

It is a one-shot instance of Kanovich's theorem that the Horn fragment of linear logic is **NP**-complete [11]: the fragment over $\{\mathsf{m}, \mathsf{fw}, \mathsf{sy}\}$ is one-shot affine Horn logic with clauses $a \multimap b$ and $a \otimes b \multimap c$, and single-output clauses already suffice. Fan-out is not needed because the input pays in advance, one resource per literal occurrence, and consistency is enforced by *consumption* rather than broadcast.

The dichotomy extends along the atoms that synchronise.

atom	why it is a join	status
sy	produces $\mathsf{fw}(b, c)$: $a \otimes b \multimap c$	proved, checked
cons_A, arity ≥ 2	consumes n messages at once	immediate
bl, br	produce a forwarder at a payload	sketched
ev	a payload carrying $\mathsf{fw}(b, c)$ releases an edge	sketched

For bl, br and ev the encoding must carry the variable's choice in payloads, since forwarding preserves a payload; the gadget is routine but not yet written out.

> *The hardness of universal negatives lives in synchronisation — the tensor on the left of the Horn arrow. Duplication alone adds none.*

Evaluation and multi-input construction are not exceptions to this: they are hard exactly because they can release or perform a join.

5.4 Where the cost of an experimenter lives: sharing

The other half of the picture is the experimenter's. Asperti proved that every simply typed λ-term normalises in a number of Lévy *families* of reductions polynomial in its size, and concluded that $\mathbf{P} = \mathbf{NP}$ "up to sharing" [1].

- *The optimal root.* η-expanding each shared higher-type variable once, before reduction, leaves every sharing node (fan) of Lamping's graphs [12] at atomic type, so fans never meet a λ or an application. The number of β-steps is then bounded by the size of the graph, all logical steps can fire before any structural one, and what remains is fan duplication and annihilation.

- *SAT.* The encoding is deterministic enumeration: a step function iterated 2^n times by n-fold composition of the Church numeral *two*. Up to sharing it costs a number of families linear in the size of the formula and quadratic in n.

- *Where the cost goes.* On the BOHM implementation, for the family P_n of conjunctions of n distinct atoms (a trivially satisfiable test family), the β-reductions of the optimal root grow as $10n^2 + 37n + 38$, while total interactions, including the oracle, go $374, 1453, 4549, 15206, 55113, 208549$ for $n = 1, \ldots, 6$ [1].

- *The corollary.* If $\mathbf{P} \neq \mathbf{NP}$, the overhead of optimal sharing cannot in general be polynomial in the number of families [1, Prop. 5.1]; compare the non-elementary lower bound of [3].

In linear-logic terms the split is between *logical* work (cuts, here communications) and *structural* work, and among the structural rules the cost is in contraction commuting with contraction

across box levels, together with the bookkeeping — Lamping's oracle — that tells paired fans from unpaired ones. On our reading, exchange and weakening contribute nothing.

Rho needs no optimal root to make contraction cheap. A doubling process, receiving a quoted term and sending back two copies, makes 2^d copies in d communications. So:

> *Hypothesis 2.9 is Asperti's Proposition 5.1 read in the other direction. A cost model that charged communications, or shared bytes, instead of unshared work would be a family-counting model in disguise, and if Conjecture 6.5 holds, Theorem 3.19 fails in it.*

This is a practical point as well as a theoretical one: content-addressed channels in cost-accounted rho are maximal sharing of code, and the phlogiston charged for content-addressed data must count unshared structural work or the metering is family counting.

6 Combining the tools

6.1 Regimes as points of a product lattice

Sections 4 and 5 separate two questions the standard formulation runs together: which atoms the *specimen* is built from, and which atoms the *experimenter* may use on it.

The specimen axis. By Proposition 5.1 and Theorem 5.2, ABSENCE is in **NL** for specimens without a synchronising atom, and **coNP**-complete as soon as sy (or a constructor of arity at least two) is present; for bl, br and ev hardness is sketched. The specimen axis measures what a *reflective* scientist must overcome. It is independent of the interaction lower bound: routing race i to stage $i+1$ only on outcome σ_i gives a join-free family with the same 2^d bound, on which ABSENCE is in **NL** for anyone who can read the specimen. The interaction bound is about access; the specimen axis is about synchronisation.

The experimenter axis. The tiers of Definition 3.11 are points:

tier	experimenter's atoms	status
linear	ev, no dd	one copy (Cor. 3.17)
interaction	ev, dd	2^d copies (Thm. 3.16)
interaction with construction	$\mathsf{ev}, \mathsf{dd}, \mathsf{cons}_A$	see below
reflection	$\mathsf{ev}, \mathsf{dd}, \mathsf{cons}_A, \mathsf{uncons}_A$	iff $\mathbf{P} = \mathbf{NP}$ (Thm. 3.19)

Name closure turns the black-box clause "no name guessing" from a stipulation into a consequence.

Proposition 6.1 (Construction does not reach down)**.** *If every internal name of the specimen contains an unforgeable component outside the closure of the experimenter's names under its constructor shapes, an experimenter over $\{\mathsf{ev}, \mathsf{dd}, \mathsf{cons}_A\}$ never sends or receives on an internal name of the specimen, and Theorem 3.16 applies to it.*

Proof sketch. By name closure every name the experimenter can produce lies in the closure of its own names and $@E$ under its constructor shapes. Constructors build upward from names present; none descends into $@E$, and ev descends only by running. An internal name with an unforgeable component outside that closure is therefore never produced. □

The adversary of Theorem 3.16, with a demonic scheduler, goes through for such an experimenter. Assuming Theorem 3.19 transfers to combinator terms, with Hypothesis 2.10 read for uncons, **P** versus **NP** in this frame is the question whether adding uncons at the top of the experimenter's axis buys more than a polynomial factor against specimens built from $\{\mathsf{m}, \mathsf{sy}\}$.

6.2 Knockouts on the combinator image

In combinator form a prefix-deleting knockout deletes an atom, and a specimen over $\{\mathsf{m}, \mathsf{fw}, \mathsf{sy}\}$ is a one-shot Horn program. A knockout therefore deletes a rule from the program, and an intervention is a set of deletions: knocking out $\mathsf{fw}(b_i, e_i^1)$ is the intervention $\sigma_i = 0$. This gives the experimental-design problem of §4 a proof-theoretic reading. Certifying that ok is not derivable is certifying non-provability of a Horn sequent, and a knockout experiment is a case split on which rules a derivation may use.

Two facts shape the design space. Knocking out the losing branch of every race leaves a confluent term, so the races are the only places an experiment needs to *choose*. And what makes the remaining question hard, by Proposition 5.1 and Theorem 5.2, is synchronisation, not the number of choices: a join-free specimen with many races is easy for a reader. This suggests measuring the hardness of a specimen by its joins rather than its races.

Conjecture 6.2 (Joins as the parameter)**.** Absence *on specimens over* $\{\mathsf{m}, \mathsf{fw}, \mathsf{sy}\}$ *is fixed-parameter tractable in the number of* sy *atoms.*

Fixing the set of joins a derivation uses turns the rest into a unit-capacity flow problem with prescribed demands at join inputs, which is polynomial. What we do not yet control is circular support, where the output of one join feeds another's input and vice versa through shared forwarders. If the conjecture holds, the hardness of universal negatives is measured exactly by how much a specimen synchronises.

6.3 Deletion versus weakening: what a grade may do

A knockout in fuzzy rho is applied in transit by the communication rule, and the experimenter controls it only by choosing a channel. That makes it tempting to place every fuzzy experiment in the interaction tier. What a grade may *do* to the code decides whether that placement is right.

Stamping and prefix deletion are interventions. The stamp $z \mapsto @(S \mid \overline{z})$ is injective (Lemma 4.4), and a prefix-deleting grade is run-subset (Lemma 4.2): neither can make a copy succeed where the specimen could not.

Proposition 6.3 (Interventions keep the bound)**.** *Theorem 3.16 holds when the interventions of Definition 3.15 are any stamping and prefix-deleting grades.*

Proof. By Lemma 4.2, a copy of E_σ that succeeds does so by a run of E_σ, which resolves every race as σ does. Under the adversary scheduler any race the copy leaves open is resolved against σ, and deleting a winner of σ kills success. So a copy succeeds on E_σ only if its grade deletes every loser of σ among the race listeners and none of its winners, so distinct σ need distinct copies; and every copy fails on $E_\perp$, whose join waits on z. The counting argument of Theorem 3.16 then applies unchanged. □

Weakening is not. A grade that drops a binding from a join, or that collapses two channels into one, relaxes a synchronisation. It creates runs the specimen does not have, and that is exactly what defeats the interaction bound.

Proposition 6.4 (Weakening defeats the interaction bound)**.** *In the family of Theorem 3.16, each of the following distinguishes $E_\perp$ from every E_σ in one experiment:*

(a) a grade that drops every binding on an e-channel from the final join and spares the binding on z;

(b) a grade that collapses e_i^0 with e_i^1 for every i and keeps z distinct.

Proof. (a) The join of E_σ becomes empty and fires at once; the join of $E_\perp$ still waits on z. (b) Each race delivers on the merged channel whichever way it resolves, so E_σ's join fires under every resolution, while the binding on z still blocks $E_\perp$. □

Weakening does not help in general. On the satisfiability specimen P_φ, dropping clause bindings from the final join, or merging t_{ij} with f_{ij}, makes every formula look satisfiable, and the experiment learns nothing. That is the expected shape: interaction lower bounds are oracle bounds, and weakening exploits the structure of particular oracles. But it locates precisely where fuzzy rho can be more than an apparatus. A weakening grade lets one run stand for many resolutions; it is a graded aggregate, the fuzzy form of the sharing of §5.4, and it must be priced as sharing. The combinator image makes the line literal: deleting an atom is run-subset, while replacing $\mathsf{sy}(a, b, c)$ by $\mathsf{fw}(b, c)$ — dropping the requirement at a — is weakening, and Theorem 5.2 says that synchronisation is exactly where the hardness lives.

Selectivity is reflection. Independently of what a grade does, how it chooses matters. A grade that selects pieces by the structure of their continuations, or merges channels by their shape (as the attenuator's meet $\overline{a} \wedge \overline{z}$ does), computes on the specimen's code. The experimenter never inspects anything, but the comparison does so on its behalf, and such a grade belongs on the reflection side of the ledger. Proposition 6.3 grants structurally selective interventions anyway, which only strengthens the bound; the line that matters for the bound is between run-subset grades and weakening ones.

6.4 Stamps as Lévy labels

Lévy's labels record the history of a redex, and two redexes are in the same family when they are copies of one another up to that history [13]. The accumulator stamp records the matched surfaces of each communication in every channel of the transmitted term, and stamps nest, $@(S' \mid S \mid \overline{z})$, once per level. It is a history label of the same kind. Communications in different stamped copies that differ only in their stamps are therefore natural candidates for a family.

Conjecture 6.5 (Rho is polynomial up to sharing)**.** *Under a cost model that charges one unit per family of stamped communications,* Absence *is decided in polynomially many families: double the specimen d times, applying at level i the two settings of race i, and read the pool through one collapsed readout.*

If Conjecture 6.5 holds, Theorem 3.19 says that, unless $\mathbf{P} = \mathbf{NP}$, no implementation of family-shared stamped execution has overhead polynomial in families — the rho form of Asperti's Proposition 5.1. The overhead has a concrete home: keeping 2^d copies apart costs stamp size, about $2^d \cdot d$ in total, and Hypothesis 2.9 charges for exactly that. The bookkeeping that Lamping's oracle performs reappears as the growth of stamped names.

6.5 A light namespace logic: the cost hypothesis as a type

Hypothesis 2.9 is a stipulation about the runtime. Light logics show how to make such a stipulation a property of the program: in light affine logic the functions on binary words typable as $!W \multimap \S^k W$ are exactly the polynomial-time ones [9, 2], and polynomiality is obtained by restricting contraction under ! through stratification. The two Asperti results are mirror images: unrestricted contraction with hidden cost gives $\mathbf{P} = \mathbf{NP}$ up to sharing; stratified contraction with honest cost gives exactly $\mathbf{P}$.

Rho is already stratified. By Proposition 2.3 a process cannot contain its own name, so quote depth is a built-in level, and by Proposition 2.4 communication raises it at most one level at a time. The leak is the duplicator, which contracts *without* climbing. Since Meredith–Radestock is a recipe for generating a namespace logic from a term language rather than a single logic [20], a light version is a choice of parameters:

- *levels are quote depth*, which rho already supplies;
- *contraction must climb*: a communication that copies a received name must place the copies one level up, so $x!(*y)$ in the duplicator does not type and a duplicating process must wrap what it sends;
- *depth is bounded independently of the input*: $[\![\Phi]\!]$ is typed at constant depth k.

In combinator terms, we expect the discipline to bound the number of alternations of dd and ev. Doubling then fits at most k rounds, and Asperti's two^n is exactly the input-dependent depth the discipline forbids.

Conjecture 6.6. *An experimenter typable in the light namespace logic at depth k runs in phlogiston polynomial in the size of its input, with degree depending on k, under any cost model that charges communications.*

Under Conjecture 6.6, together with the converse — every polynomial-phlogiston reflective experimenter is equivalent to one of light type, as light affine logic is complete for polynomial time — Theorem 3.19 has a typed form: $\mathbf{P} = \mathbf{NP}$ if and only if some reflective experimenter of light type decides Absence on specimens over $\{\mathsf{m}, \mathsf{sy}\}$ — that is, certifies non-derivability in one-shot Horn logic. The restatement is still tautological, but the cost model has moved from the runtime into a judgement that can be checked. Light affine logic also restricts boxes to at most one free assumption; whether the rho analog — a quoted process sent at a level mentions at most one name from the level below — is natural, or too strong to express the specimens at all, is the first thing to test.

6.6 The barriers, located

The correspondences in this subsection are analogies; we record them because they say where in the apparatus each barrier would bite. *Relativisation* is interaction: arguments uniform over interaction regimes treat the specimen as an oracle and can separate unconditionally (Theorem 3.16) while saying nothing about reflection (Remark 3.18). *Aggregation* is collapse: graded readouts that identify resolutions are the apparatus in which an analog of algebrisation would sit, and Proposition 6.4 shows they must be costed by sharing, not by communication. *Natural proofs* would be experiments that certify hardness through a large, constructive property of the table of outcomes over all knockouts [26]; making that precise is open.

7 Living in either world

Three cautions frame what follows. First, **P** versus **NP** is a worst-case question, and a mortal scientist lives in a distribution over ecologies [17, §"Distributions over ecologies"]; what it experiences is an average over that distribution, which is the territory of Impagliazzo's five worlds [10]. Second, polynomial is not affordable: the budget is finite in both worlds, and what changes is how the price of a universal negative scales with the size of the specimen. Third, only reflection is affected. Interaction regimes pay exponentially in both worlds (Theorem 3.16). The difference between the worlds is mainly a difference in what it is worth to *hold* a specimen, and in what it costs to find one.

7.1 A mortal scientist in a world where P = NP

The asymmetry survives in logic and collapses in price. The scientist's evidence is still asymmetric: a universal is still never confirmed by running. But a reflective scientist holding a bounded specimen can certify its universal negatives for a price polynomial in its size, as it already could confirm an existential. Popper's asymmetry becomes a fact about interaction only.

Holding is everything. Because only reflection gains, acquiring a specimen's code — not its stack, which is what a harvest takes, but a quoted copy of its program — becomes the decisive act. The pressure recorded in [17, Prop. Scientists are preferred prey] and in its discussion of theory theft sharpens: scientists, whose hypotheses are now cheap to check once held, become even more strongly preferred prey. The destructive assay still forces induction — one specimen, one measurement — but that one measurement now yields complete bounded predictions for every member of the specimen's kind that shares its code.

Concealment fails on two fronts. Concealment by depth fails once a specimen is held, since structure is cheap to certify. Concealment by unforgeability weakens too: deriving a name that satisfies a perceivable predicate is a search problem with a checkable witness, so capability gating loses its force, and implementations whose unforgeable names rest on one-way functions lose them outright. Only the structural freshness of Proposition 2.3 survives. Foraging becomes cheap, the profitable band widens, and predation pressure rises.

Imperfection must come from elsewhere. The companion game note found that a solved science is a dead ecology: a monoculture of perfect players draws every game and starves [8]. In a world where **P** = **NP**, every bounded question with short certificates — including every universal negative about a bounded specimen — is decided polynomially by whoever holds the code. Games of polynomial length are **PSPACE**-hard and stay out of reach unless **P** = **PSPACE**, but the imperfection that keeps an ecology alive can no longer come from the hardness of what can be checked. It must come from budgets, and from the fact that most of what a scientist meets is met across a cut, by interaction, where the exponential remains. The ecology stays alive where specimens cannot be held.

7.2 A mortal scientist in a world where P ≠ NP

Universal negatives have a price floor. Even with the code in hand, certifying that a bounded specimen cannot do something costs, in the worst case, more than any polynomial.

The scientist lives with $\perp$ and with universals that are only ever refuted. Its theories stay defeasible, and the budget remains its inductive bias with complexity reinforcing it rather than replacing it.

Synchronisation is camouflage. By Proposition 5.1 and Theorem 5.2, a specimen that does not synchronise is transparent to a reflective scientist: what it cannot do is cheap to know once it is held. A specimen that races and joins can hide what it cannot do. In such a world coordination is a defensive adaptation, and a scientist does well to specialise on niches whose inhabitants synchronise little, where its universals are cheap. The search for science is then, in part, the search for tractable fragments — the light, stratified environments of §6.5 — in which universal negatives can be afforded.

Science does not complete. Intrinsic hardness supplies the imperfection the game note found load-bearing [8], so the ecology does not run down to a solved monoculture. Concealment is a price rather than a wall, but at scale the price is real; and whether capability gating holds depends on more than $\mathbf{P} \neq \mathbf{NP}$, namely on whether one-way functions exist [10].

Rational epistemics. The scientist should invest in existential hypotheses, which are cheap to confirm, and hold universals as priced bets. Knockouts, controls and pooling cut costs by structure and by constants; they do not escape the exponent. The value of reflection is real but bounded: reading a specimen buys universals only for specimens that live in tractable fragments.

The ecology is the adversary. Finally, the worst-case caveat may matter less than it seems. Theorem 3.16 is an adversary argument, and an ecology supplies an adversary: selection favours prey whose universal negatives are expensive to certify, since those are the prey whose defences cannot be cheaply ruled out. If that pressure is strong enough, the distribution a mortal scientist actually meets drifts toward hard specimens, and in an ecology the worst case is not exceptional but evolved. We state this as a conjecture about ecologies rather than about complexity classes: *predation selects for synchronisation.* $\mathbf{P} \neq \mathbf{NP}$ alone does not make hard specimens easy to come by — in Impagliazzo's Heuristica they are rare — but where they can be generated, that selection would produce an opaque world.

8 Open problems

1. **Joins as the parameter.** Prove or refute Conjecture 6.2; the obstacle is circular support through shared forwarders.
2. **Reading by guards.** Write out the reconstruction of a held specimen by adaptive structural guards used in Theorem 3.19.
3. **The remaining joins.** Write out the payload-routing gadgets showing **coNP**-hardness over $\{\mathsf{m}, \mathsf{bl}\}$, $\{\mathsf{m}, \mathsf{br}\}$ and $\{\mathsf{m}, \mathsf{fw}, \mathsf{ev}\}$.
4. **Equivariant stamping.** Realise (E3) for general rho processes by a namespace-relative evaluation in which every name the copy synthesises is stamped.
5. **Drop-only extensionality.** Prove Conjecture 3.12, making the interaction tier an admissible observer class in the sense of [28].

6. **Grades.** Run-subset grades (stamping, prefix deletion) preserve Theorem 3.16; weakening grades (dropped bindings, collapse) defeat it on particular families. Characterise the weakening grades that help on hard specimens, if any, and price them by sharing.

7. **Models.** Relate the deletion model of Construction 4.1 to the accumulator and attenuator models of [7], and to the lift of fuzzy rho to ci GSLTs with binding.

8. **Families of stamped communications.** Define them, and prove or refute Conjecture 6.5.

9. **A light namespace logic.** Instantiate the recipe of [20] with stratified contraction, and prove Conjecture 6.6; decide whether the one-assumption restriction is needed.

10. **The ecology as adversary.** Test, in the ensemble of [17], whether predation selects for synchronising prey.

11. **Cost hypotheses.** Verify Hypotheses 2.8 to 2.10 for cost-accounted rho as implemented, with particular attention to content-addressed sharing.

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A Status of results

Result	Status	Note
Prop. 2.3	known	[17, 15, 14]
Prop. 2.4	proved	extends [15] to joins
Prop. 3.3	proved	
Constr. 3.5 (adequacy)	sketched	administrative phases as in [15]; observer's message consumed
Thm. 3.7, Prop. 3.8	proved	
Lem. 3.9	proved	syntactic
Conj. 3.12	conjecture	open problem 5
Lem. 3.14	proved	relative to Hyp. 2.9, Def. 3.13
Thm. 3.16, Cor. 3.17	proved	relative to the outcome model of Def. 3.15 (granted interventions and quiescence); cost via Hyp. 2.9
Thm. 3.19, Cor. 3.20	proved	relative to Hyps. 2.8–2.10; reading by guards sketched; a restatement
Constr. 4.1	proved	quantale, action, comparison and (K1)–(K3); order axiom with μ into names, as in the minimal model
Lem. 4.2	proved	prefix pieces not beneath a lift
Lem. 4.4, 4.5	proved	
Ex. 4.6	proved	
Prop. 5.1	proved	
Thm. 5.2	proved	checked over $\{\mathsf{m}, \mathsf{fw}, \mathsf{sy}\}$ on 920 random CNFs
Joins table: cons_A	immediate	
Joins table: $\mathsf{bl}, \mathsf{br}, \mathsf{ev}$	sketched	open problem 3
Prop. 6.1	sketched	from name closure [18]; adversary with demonic scheduler
Prop. 6.3, 6.4	proved	
Conj. 6.2, 6.5, 6.6	conjecture	open problems 1, 8, 9
§6.6, §7	analogy, discussion	not theorems